

2. LITERATURE SURVEY

2.1 Background

Plant disease identification is a complex and critical aspect of agriculture. Historically, farmers have depended on physical inspection to detect visible signs of disease. While experts can often recognize the symptoms accurately, this method is impractical for large-scale farming and is highly subjective.

The emergence of computer vision and deep learning provides a scalable solution for this issue. Models such as Convolutional Neural Networks (CNNs) have demonstrated significant success in image classification tasks, including medical image analysis, object detection, and, more recently, agricultural disease detection.

With the advancement in mobile devices and Internet connectivity, AI models can now be deployed closer to the point of need, empowering farmers even in remote regions to access smart diagnostic tools.

2.2 Existing Systems

Several researchers have proposed different methods for plant disease detection:

1. "Plant Leaf Disease Detection Using Convolutional Neural Network"
— Journal of Artificial Intelligence (2020)
This paper proposed the use of CNN for classifying plant leaf images.

Their system achieved an accuracy of 96% on a custom dataset and proved the efficiency of deep learning over traditional methods.

2. "Mobile Plant Disease Identification System Using Deep Learning" —IEEE-Access(2021)

This study introduced a mobile application that uses a pre-trained MobileNet model for disease identification. It stressed the importance of lightweight models suitable for real-world deployment.

3. "Deep Learning-Based Plant Disease Detection: A Comprehensive Review" — Computers and Electronics in Agriculture (2022)

The paper reviewed over 100 studies and highlighted that CNN architectures like ResNet, VGG, and DenseNet perform exceptionally well on plant disease datasets.

4. "Detection of Banana Leaf Diseases Using Transfer Learning" — International Journal of Computer Applications (2023)

The researchers specifically worked on banana leaf diseases using pre-trained models and reported around 94% accuracy, emphasizing the need for a localized dataset for better performance.

5. "AI in Agriculture: Current Trends and Future Prospects" — Springer-Nature(2022)

The paper discusses how AI is revolutionizing agriculture, covering crop monitoring, disease detection, and yield prediction.